Cheon's attack on curves shaped like P-224: measured scaling and extrapolation

Study `research/cheon_p224_shape_20261022/`. Protocol (pre-registered, committed before any measured run): PROTOCOL.md. Run report (observations only): reports/run-20261022T0538Z/RUN_REPORT.md. Raw records: `runs/`, `sessions/`. Code: `src/cryptanalysis/cheon_counted.rs`, `examples/cheon_p224_shape*.rs`, `tests/cheon_counted.rs`.

This is the **DLP with auxiliary inputs**, not the plain ECDLP. The Cheon arms are handed `[alpha^i]G` that rho cannot use. No row here is an ECDLP advance, and no index-calculus baseline is changed.

1. Question and method in one paragraph

NIST P-224 has `n - 1 = 2^2 3^6 5 17 2153 q195` (smooth part `d- = 2^28.99`) and `n + 1 = 2 7 19 641707280681 q177` (`d+ = 2^47.28`). The formulas give Cheon costs `sqrt(n/d) + sqrt(d) = 2^97.50` (`n - 1`) and `sqrt(n/d) + d = 2^88.36` (`n + 1`, about `2d` leaked powers) against rho at `2^111.83`. We built prime-order curves of 32 to 56 bits (`n` has 32, 36, ..., 56 bits) whose `n - 1` (resp. `n + 1`) equals `d * q` with `q` prime and `log d / log n` within 0.015 of P-224's ratio (0.12943, resp. 0.21106). **Construction:** random `(a, b)` from a SplitMix64 stream, `#E` by baby-step giant-step over the Hasse interval (`O(p^(1/4))`), first curve in stream order with prime `n` and the shape (`PROTOCOL.md` 2.2); no CM. Re-verified by `cheon_p224_shape check` ([n]G = O, Hasse interval, primality, shape). Then, on the same 6 planted targets per curve: native counted Cheon `n - 1` (fixed-base comb, and the naive double-and-add algorithm of `cheon_attack.rs`), native counted Cheon `n + 1` (norm-one torus of `F_(n^2)`, comb), and ecbench's `rho.negation`. Every run's recovered logarithm was checked against the planted one. Counts are in the repository's unit (one affine addition or doubling), every phase charged, unpriced native work listed and priced at 0, 0.1 and 1 group-addition equivalents.

2. The frozen curves

family	bits	n	log2 n	d	log2 d / log2 n	leaked points used	highest power index needed
minus	32	3221255537	31.58	16	0.1266	2	16
minus	36	51539573731	35.58	30	0.1379	2	30
minus	40	824632004309	39.58	28	0.1214	2	28
minus	44	13194139832747	43.58	74	0.1425	2	74
minus	48	211106219695673	47.58	56	0.122	2	56
minus	52	3377699780661797	51.58	164	0.1426	2	164
minus	56	54043195377297631	55.58	210	0.1388	2	210
plus	32	3221175689	31.58	90	0.2055	180	180
plus	36	51539736719	35.58	240	0.2222	480	480
plus	40	824633358973	39.58	398	0.2182	796	796
plus	44	13194137032873	43.58	574	0.2103	1148	1148
plus	48	211106244944429	47.58	990	0.2091	1980	1980
plus	52	3377699707176467	51.58	1836	0.2102	3672	3672
plus	56	54043195250336543	55.58	2016	0.1975	4032	4032

`d` and `n` are exact; the ratio differs from P-224's by at most 0.015 by construction. The `n - 1` variant consumes 2 leaked points (`[alpha]G`, `[alpha^d]G`), but a protocol that publishes sequential powers must publish up to index `d` to include `[alpha^d]G`; the `n + 1` variant consumes the `2d` points `[alpha^i]G`, `i <= 2d`. At the P-224 shape these are `2^29.0` and `2^48.3` sequential powers.

3. Observations (measured)

Every run below recovered the planted logarithm: 126 Cheon runs and 84 rho runs, 0 failures, 0 wrong answers, 0 timeouts (Table 1b). Full tables with phases and unpriced counters are in the run report; the table that carries the requested columns (size, arm, ops, `S`, ratio to rho, leaked powers, correct) is Table 1. `ops` columns are medians over the 6 verified targets; `S = ops / sqrt(n)`; "ratio to rho" is the median over targets of the paired ratio on the same target, "ratio of sums" is the ecbench-style totals ratio; "leaked powers" is the number of leaked points the arm consumes.

Table 1. Measured operation counts, every run

Unit: group operations (one affine addition or doubling = 1), every phase charged. `S = median ops / sqrt(n)`. `F(n,d)` is the formula `sqrt(n/d)+sqrt(d)` (minus) or `sqrt(n/d)+d` (plus). `ratio to rho` is the median over targets of the paired ratio (same curve, same target). `correct` counts recovered logarithms equal to the planted one.

family	bits	n	d	arm	leaked powers	correct	median ops	min	max	S	ops/F	ratio to rho (median of pairs)	ratio of sums
minus	32	3221255537	16	rho.negation	0	6/6	36,521	19,231	72,960	0.643	2.6	1 (reference)	1
minus	32	3221255537	16	cheon.pminus1_comb	2	6/6	56,960	42,502	68,568	1.004	4.0	1.31	1.45
minus	32	3221255537	16	cheon.pminus1_naive	2	6/6	986,396	665,226	1,243,458	17.380	69.5	22.21	25.18
minus	36	51539573731	30	rho.negation	0	6/6	253,101	150,301	522,990	1.115	6.1	1 (reference)	1
minus	36	51539573731	30	cheon.pminus1_comb	2	6/6	159,115	118,228	179,279	0.701	3.8	0.66	0.56
minus	36	51539573731	30	cheon.pminus1_naive	2	6/6	3,377,689	2,347,739	3,885,036	14.878	81.5	14.12	11.79
minus	40	824632004309	28	rho.negation	0	6/6	444,282	71,008	1,447,215	0.489	2.6	1 (reference)	1
minus	40	824632004309	28	cheon.pminus1_comb	2	6/6	818,743	552,699	935,245	0.902	4.8	1.70	1.32
minus	40	824632004309	28	cheon.pminus1_naive	2	6/6	15,237,704	10,234,088	17,428,404	16.780	88.8	31.68	24.64
minus	44	13194139832747	74	rho.negation	0	6/6	3,285,487	2,167,521	4,683,393	0.905	7.8	1 (reference)	1
minus	44	13194139832747	74	cheon.pminus1_comb	2	6/6	2,004,198	1,549,214	2,431,239	0.552	4.7	0.61	0.61
minus	44	13194139832747	74	cheon.pminus1_naive	2	6/6	41,320,964	31,860,597	50,204,230	11.376	97.9	12.58	12.56
minus	48	211106219695673	56	rho.negation	0	6/6	21,834,523	2,843,639	32,193,596	1.503	11.2	1 (reference)	1
minus	48	211106219695673	56	cheon.pminus1_comb	2	6/6	7,285,384	6,051,895	8,837,789	0.501	3.8	0.35	0.36
minus	48	211106219695673	56	cheon.pminus1_naive	2	6/6	165,247,846	137,141,584	200,622,954	11.373	85.1	7.89	8.06
minus	52	3377699780661797	164	rho.negation	0	6/6	61,986,494	39,293,404	120,295,824	1.067	13.7	1 (reference)	1
minus	52	3377699780661797	164	cheon.pminus1_comb	2	6/6	21,471,423	18,487,349	30,042,391	0.369	4.7	0.32	0.33
minus	52	3377699780661797	164	cheon.pminus1_naive	2	6/6	399,345,336	343,790,499	558,913,697	6.871	88.0	5.90	6.17
minus	56	54043195377297631	210	rho.negation	0	6/6	158,861,495	134,220,877	231,679,321	0.683	9.9	1 (reference)	1
minus	56	54043195377297631	210	cheon.pminus1_comb	2	6/6	100,364,872	82,959,549	122,760,558	0.432	6.3	0.59	0.60
minus	56	54043195377297631	210	cheon.pminus1_naive	2	6/6	2,017,940,940	1,667,841,393	2,468,411,403	8.680	125.8	11.94	12.02
plus	32	3221175689	90	rho.negation	0	6/6	51,436	27,244	93,796	0.906	8.5	1 (reference)	1
plus	32	3221175689	90	cheon.pplus1_comb	180	6/6	95,294	58,721	113,909	1.679	15.7	1.82	1.59
plus	36	51539736719	240	rho.negation	0	6/6	187,614	114,238	346,591	0.826	12.6	1 (reference)	1
plus	36	51539736719	240	cheon.pplus1_comb	480	6/6	222,988	150,802	248,849	0.982	15.0	1.08	1.00
plus	40	824633358973	398	rho.negation	0	6/6	1,087,652	214,874	2,075,309	1.198	23.7	1 (reference)	1
plus	40	824633358973	398	cheon.pplus1_comb	796	6/6	704,630	388,117	941,678	0.776	15.3	0.70	0.55
plus	44	13194137032873	574	rho.negation	0	6/6	4,150,295	1,490,804	4,632,503	1.143	27.3	1 (reference)	1
plus	44	13194137032873	574	cheon.pplus1_comb	1148	6/6	2,419,814	1,509,216	2,845,511	0.666	15.9	0.63	0.66
plus	48	211106244944429	990	rho.negation	0	6/6	10,502,984	504,689	19,888,818	0.723	22.7	1 (reference)	1

family	bits	n	d	arm	leaked powers	correct	median ops	min	max	S	ops/F	ratio to rho (median of pairs)	ratio of sums
plus	48	211106244944429	990	`cheon.pplus1_comb`	1980	6/6	5,077,816	3,626,295	9,339,109	0.349	11.0	0.63	0.59
plus	52	3377699707176467	1836	`rho.negation`	0	6/6	64,022,410	39,514,105	119,465,498	1.102	47.1	1 (reference)	1
plus	52	3377699707176467	1836	`cheon.pplus1_comb`	3672	6/6	22,508,339	12,957,180	35,433,734	0.387	16.6	0.33	0.34
plus	56	54043195250336543	2016	`rho.negation`	0	6/6	429,577,620	84,161,122	519,640,803	1.848	82.9	1 (reference)	1
plus	56	54043195250336543	2016	`cheon.pplus1_comb`	4032	6/6	98,342,762	49,002,962	131,761,087	0.423	19.0	0.21	0.24

Unpriced native work (AGENTS.md 5a) is listed for every arm in Table 2 of the run report (`reports/run-20261022T0538Z/RUN_REPORT.md`), with the Cheon/rho ratio at 0, 0.1 and 1 group-addition equivalents per unpriced operation, the same weight on both arms. The order (Cheon dearer or cheaper than rho) is the same at all three weights in 20 of the 21 Cheon-versus-rho cells (7 sizes, 3 Cheon arms); the one unresolved cell is `plus` 36 bits (1.08, 1.00, 0.66). Rho's unpriced work is one canonicalisation per walk step (about as large as its charged count), so raising the weight lowers every Cheon/rho ratio.

Phases. For `cheon.pminus1_comb` the baby and giant steps of step 1 are 98.9% to 100% of the operations from 40 bits up (table construction 21.6% at 32 bits, 0.0% at 56). For `cheon.pplus1_comb` the polynomial phase (the `3(2d+1)` multi-scalar terms) is 12.8% at 32 bits and 0.5% at 56 bits; baby and giant steps are 99.5% at 56 bits (Table 6 of the run report, Fig. 3).

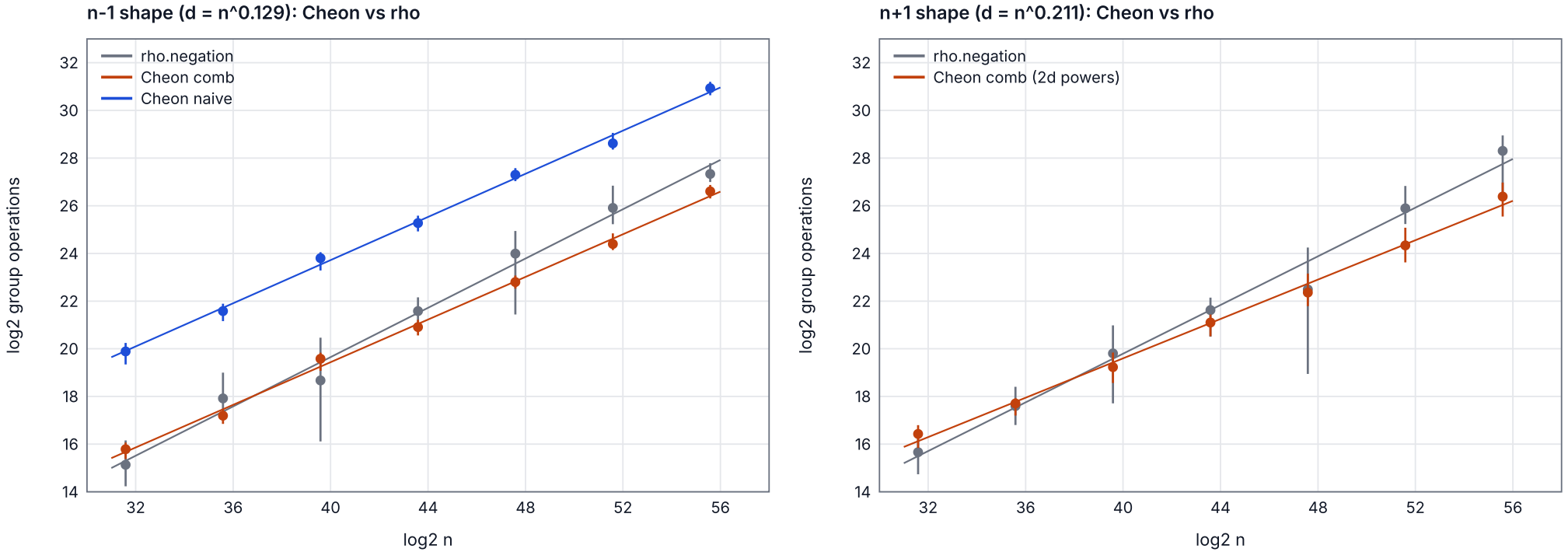


Fig. 1. MEASURED log2 operations per size (dots = mean over verified targets, bars = min..max) with fit A over the measured range. Source: results.json.

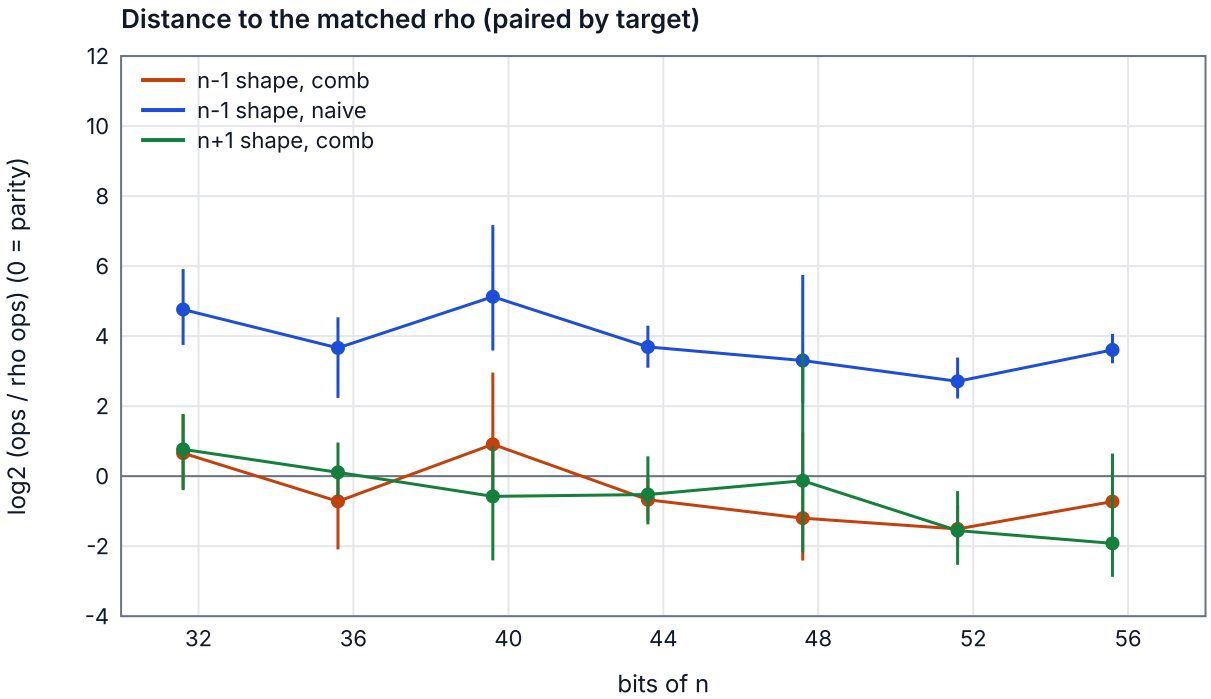


Fig. 2. log2 of the Cheon/rho operation ratio per size (dots = mean, bars = min..max over paired targets). Above 0 = Cheon dearer. Source: runs.jsonl and the ecbench rho session.

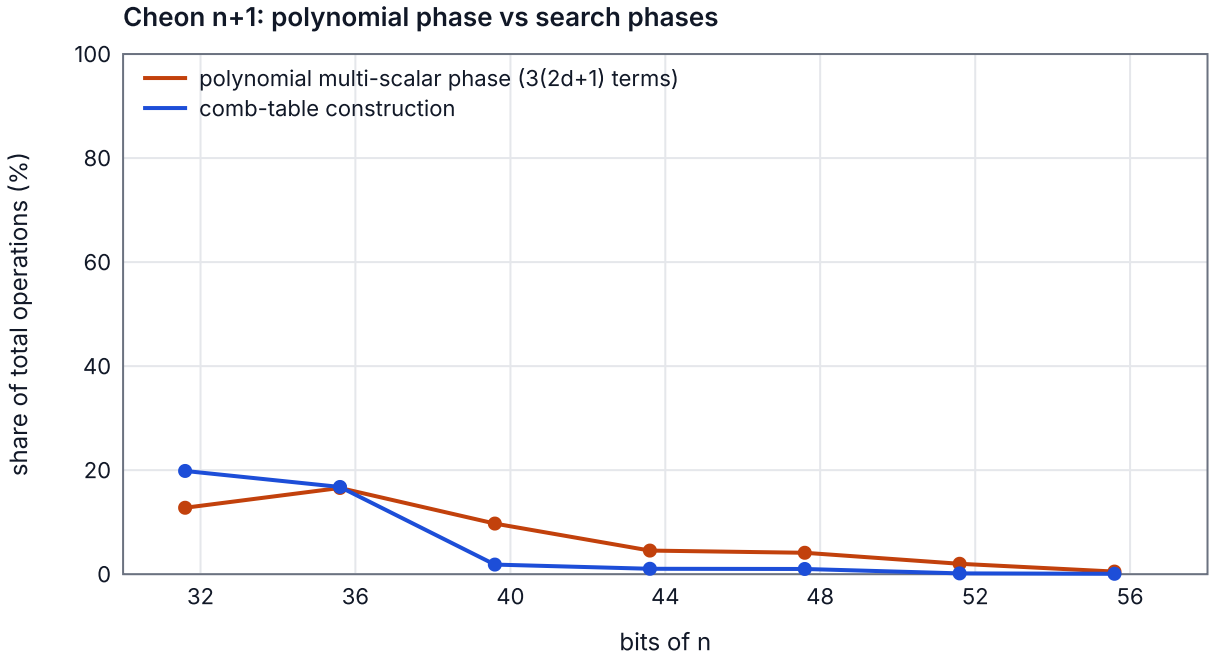


Fig. 3. Median share of the n+1 variant's operations by phase. Source: runs.jsonl phase columns.

4. Calculation (derived from section 3; no new runs)

All fits are in `results/results.md` and `results/results.json`, produced by `examples/cheon_p224_shape_analyse.rs` and recomputed independently by `examples/cheon_p224_shape_validate.rs` (`validator-report.json`). Fit A is OLS of the per-size mean of `log2(ops)` on `log2 n` over the 7 sizes; "t 95%" is the Student interval (5 degrees of freedom), "boot 95%" the percentile interval over 2000 resamples of the targets inside each size. Fit B regresses `log2(ops / F(n,d))` on `log2 n` and adds `log2 F` at the P-224 shape. Fit C (`n + 1` only) fits the polynomial phase on `log2 d` and the search phases on `log2 sqrt(n/d)` and adds the two.

Table 3. Fitted exponents, `log2(ops) = a + b * log2(n)`

Ordinary least squares on the per-size mean of `log2(ops)` over verified targets (each size one point). `t 95%` is the Student interval (sizes - 2 degrees of freedom); `boot 95%` is the percentile interval from 2000 resamples of the targets inside every size. Sizes used are listed.

family	arm	sizes	slope b	s.e.	t 95%	boot 95%	prediction (PROTOCOL 5)	inside
minus	`rho.negation`	7	0.517	0.024	[0.455, 0.579]	[0.494, 0.540]	[0.45, 0.55]	inside
minus	`cheon.pminus1_comb`	7	0.447	0.012	[0.417, 0.477]	[0.439, 0.456]	[0.40, 0.50]	inside
minus	`cheon.pminus1_naive`	7	0.453	0.011	[0.425, 0.481]	[0.444, 0.462]	none	n/a (no prediction)
plus	`rho.negation`	7	0.511	0.028	[0.438, 0.584]	[0.478, 0.540]	[0.45, 0.55]	inside
plus	`cheon.pplus1_comb`	7	0.413	0.013	[0.379, 0.447]	[0.397, 0.428]	[0.30, 0.46]	inside

Table 4. EXTRAPOLATION to log2 n = 224 (not measured)

Every figure here is an extrapolation of a fit over the sizes in Table 3, from at most 56 bits to 224 bits. Fit A is the free power law of Table 3. Fit B regresses `log2(ops / F(n,d))` on `log2 n` and adds the formula value at the P-224 shape (`F = 2^97.50` minus, `2^88.36` plus). Fit C (plus only) fits the polynomial phase on `log2 d` and the search phases on `log2 sqrt(n/d)` separately and adds them. 'pred 95%' is the OLS prediction interval; 'boot 95%' the bootstrap interval of the extrapolated value. Gap = extrapolated minus formula, in bits.

family	arm	fit	log2 ops at 224	pred 95%	boot 95%	formula	gap (bits)	S/S_rho at 224
minus	`rho.negation`	A	114.78	[103.5, 126.1]	[110.7, 118.8]	111.83	+2.96	2^+0.0
minus	`cheon.pminus1_comb`	A	101.71	[96.3, 107.1]	[100.3, 103.2]	97.50	+4.21	2^-13.1
minus	`cheon.pminus1_comb`	B (slope of log2 ops/F on log2 n: 0.023 +- 0.010)	103.86	[99.2, 108.5]		97.50	+6.36	2^-10.9
minus	`cheon.pminus1_naive`	A	107.02	[101.9, 112.1]	[105.4, 108.8]	97.50	+9.52	2^-7.8
minus	`cheon.pminus1_naive`	B (slope of log2 ops/F on log2 n: 0.029 +- 0.007)	109.17	[105.7, 112.6]		97.50	+11.67	2^-5.6
plus	`rho.negation`	A	113.78	[100.5, 127.0]	[107.8, 119.0]	111.83	+1.95	2^+0.0
plus	`cheon.pplus1_comb`	A	95.59	[89.3, 101.8]	[92.6, 98.3]	88.36	+7.23	2^-18.2
plus	`cheon.pplus1_comb`	B (slope of log2 ops/F on log2 n: 0.006 +- 0.008)	93.32	[89.4, 97.2]		88.36	+4.96	2^-20.5
plus	`cheon.pplus1_comb`	C (poly: log2 ops = 5.83 + 1.187 log2 d, search: 3.03 + 1.041 log2 sqrt(n/d))	95.01			88.36	+6.65	2^-18.8

Table 5. Distance to the matched rho (largest measured size, and EXTRAPOLATED crossing)

`ratio` is the median paired Cheon/rho ratio at the largest size with a complete arm. `crossing` is the `log2 n` at which fit A of the arm meets fit A of rho (extrapolation; `none` if the arm's slope is not below rho's).

family	arm	largest size	ratio at largest size	crossing log2 n (extrapolated)
minus	`cheon.pminus1_comb`	56	0.59	37
minus	`cheon.pminus1_naive`	56	11.94	103
plus	`cheon.pplus1_comb`	56	0.21	38

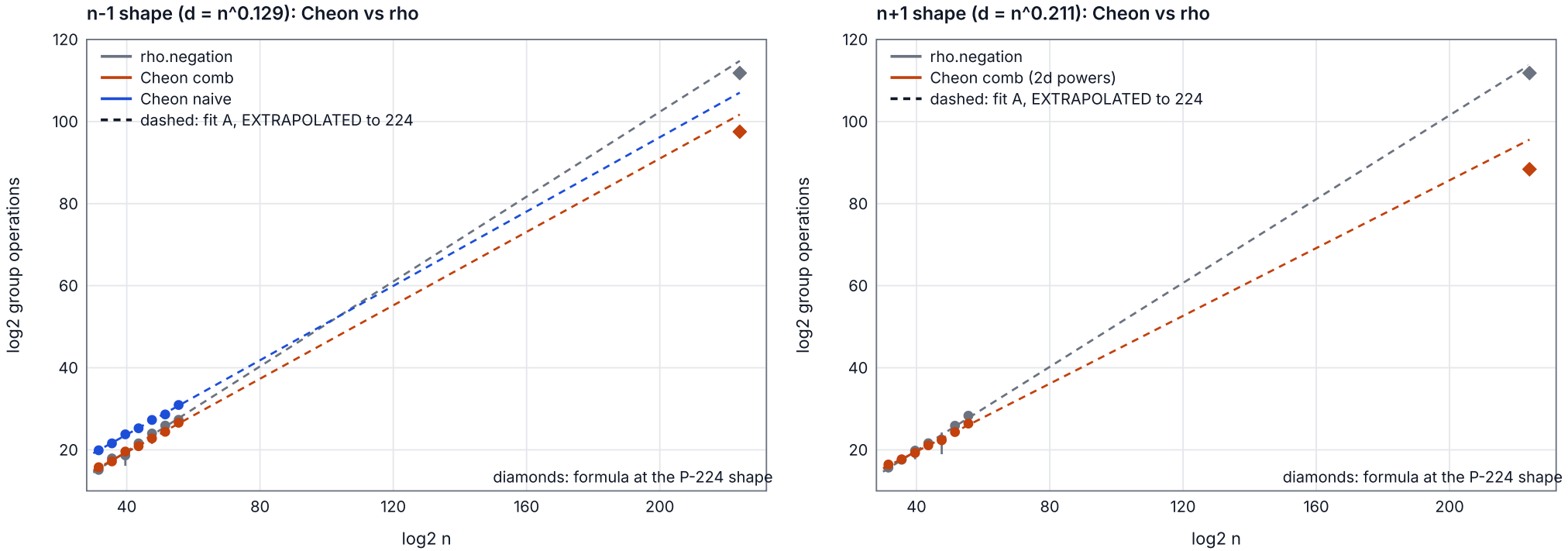


Fig. 4. EXTRAPOLATION. Fit A extended from the measured sizes to 224 bits (dashed); diamonds are the formula values at the P-224 shape. Nothing right of 58 bits is measured.

Calibration of the extrapolation against a known answer. Rho's own fit extrapolates to `2^114.78` (`minus` curves) and `2^113.78` (`plus` curves) at 224 bits, against the exact formula `2^111.83` (`0.886 sqrt(n)`): the method overshoots a known quantity by +2.96 and +1.95 bits, because the fitted rho slopes (0.517, 0.511) lie above 1/2. Subtracting that overshoot from the Cheon gaps of Table 4 (arithmetic: gap(Cheon) - gap(rho) on the same family and fit A) leaves `+4.21 - 2.96 = +1.25` bits (`minus`, comb) and `+7.23 - 1.95 = +5.28` bits (`plus`, comb) above the formulas `2^97.50` and `2^88.36`.

5. Pre-registered predictions against outcome (PROTOCOL.md section 5)

id	prediction (written before data)	outcome	verdict
H1	`minus` Cheon comb slope in [0.40, 0.50]	0.447, t 95% [0.417, 0.477], boot [0.439, 0.456]. Formula alone gives `(1 - 0.129)/2 = 0.435`	inside
H2	`plus` Cheon comb slope in [0.30, 0.46]	0.413, t 95% [0.379, 0.447], boot [0.397, 0.428]. Formula alone gives 0.395	inside
H3	rho slope in [0.45, 0.55] and per-size `S_rho` in [0.6, 2.0]	slopes 0.517 and 0.511 (t 95% [0.455, 0.579], [0.438, 0.584]); per-size median S 0.49 to 1.85 (0.49 at 40 bits `minus` is below 0.6)	slope inside; S band missed once at the median level
H3 gate	rho unreliable if slope outside [0.40, 0.60] or "any rho S outside [0.3, 4]"	slope gate met. **5 of 84 individual rho runs have S below 0.3** (minimum 0.035, maximum 2.30); per-size medians are inside [0.3, 4]	literal reading of "any" tripped; see section 7, deviation 1
H4	slope of `log2(ops/F)` on `log2 n` within +-0.08	`minus` comb 0.023 +- 0.010; `minus` naive 0.029 +- 0.007; `plus` comb 0.006 +- 0.008	inside
H5	both Cheon variants dearer than rho at every size; fitted lines cross between 55 and 130 bits	`minus` comb is cheaper than rho (median paired ratio below 1) at 36, 44, 48, 52 and 56 bits and dearer at 32 and 40; `plus` comb dearer at 32 and 36, cheaper from 40 bits; fit-A lines cross at 37 and 38 bits (inside the measured range); the naive arm is dearer at	**falsified**

id	prediction (written before data)	outcome	verdict
		every size (5.9 to 31.7, crossing extrapolated to 103 bits)	
H6	extrapolation `minus` 2^101 to 2^103, `plus` 2^94 to 2^97	fit A: 2^101.71 (`minus`), 2^95.59 (`plus`); fit B: 2^103.86 and 2^93.32; fit C: 2^95.01	fit A and C inside; fit B outside by 0.86 and 0.68 bits
control	`cheon_attack.rs` agrees with `pminus1_naive`	9 of 9 identical logarithm and exponentiation count	met
control	wrong auxiliary input must not recover alpha	`tests/cheon_counted.rs` (both variants), output in `unit-tests.txt`: 6 of 6 pass	met

6. Interpretation (separate from the observations above)

Scope. These are DLPwAI measurements. Every Cheon arm was given leaked points that the rho arm cannot use; the leak is free in the cost and is reported as its own column. Nothing here lowers the plain ECDLP cost, and no index-calculus pipeline was run or changed, so there is no improvement over the index-calculus baseline to report (AGENTS.md "Success criterion").

The formulas' exponents are measured. The fitted Cheon exponents (0.447 for `n - 1`, 0.413 for `n + 1`) sit 0.012 and 0.019 above the exponents the formulas imply at these shapes (0.435, 0.395); the excess is the size of the slow log factor of an exponentiation and of the `d` term. Rho's fitted exponents are 0.517 and 0.511 against 1/2. The cost of Cheon relative to rho therefore falls as `n^(-0.07)` (`minus`) and `n^(-0.10)` (`plus`) over 32 to 56 bits (difference of the fitted slopes; the rho slope has s.e. 0.024 to 0.028). The cost relative to the DLPwAI formula `F` is flat: `ops/F` is 3.8 to 6.3 (`minus` comb), 70 to 126 (`minus` naive) and 11.0 to 19.0 (`plus`), with no significant trend (H4). The comb-versus-naive gap is a constant factor of 17 to 23 on the same exponent (slopes 0.447 and 0.453).

Classification (AGENTS.md 3). `cheon.pminus1_comb` and `cheon.pplus1_comb`: advance on the DLPwAI (ratio to the plain-ECDLP floor `sqrt(n)` falls with `n`; ratio to the DLPwAI formula is flat), not applicable to the ECDLP because the inputs are leaked powers. `cheon.pminus1_naive` against `pminus1_comb`: engineering (constant factor, same exponent). Matched rho: reference.

Distance remaining to the matched rho (not a pass or fail). At the largest size run (55.6 bits) the median paired Cheon/rho ratio is 0.59 (`minus`, comb), 0.21 (`plus`, comb) and 11.9 (`minus`, naive). At the P-224 shape the fits put the Cheon cost at about `2^101.7` to `2^103.9` (`n - 1` variant) and `2^93.3` to `2^95.6` (`n + 1` variant), i.e. 4.2 to 6.4 and 5.0 to 7.2 bits above the formulas `2^97.50` and `2^88.36`, against a rho extrapolation of `2^114.8` and `2^113.8` (known value `2^111.83`). **All of the 224-bit figures are extrapolations** over a factor of `2^168` in `n` beyond the largest measured curve, with 95% prediction intervals 11 to 13 bits wide for the Cheon fits and 23 to 27 for rho (Table 4); the calibration paragraph in section 4 shows the method's own bias on the rho reference is +2 to +3 bits. The measured constant above the formula is the distance between the formula and a counted implementation in this repository's unit: about 2^2.2 (`n - 1`, comb) and 2^4.0 (`n + 1`).

The leak is the price. At the P-224 shape the `n - 1` variant needs `[alpha^d]G` with `d = 2^29.0`; a protocol publishing sequential powers must publish `5.3 x 10^8` of them, and a static-DH oracle (Brown-Gallant) must be queried `d` times to produce it. The `n + 1` variant consumes `2d = 2^48.3` points (`3.4 x 10^14`). The measured table's "leaked powers" column grows from 180 to 4,032 over the `plus` sizes; the extrapolated `2^88` to `2^95` cost is conditional on those `2^48` powers existing. This study does not claim that any deployed P-224 protocol publishes such powers.

Against the legacy Python measurement (`research/torsion_auxiliary_inputs_20260918`, `RESEARCH_TORSION_AUXILIARY_INPUTS.md`, pure Python, different curves with `d` near `n^(1/2)` and `n^(1/3)`): the comb constants there were `15` to `34` times `sqrt(n/d)` and `125` to `185` times for `n + 1`; here they are 3.8 to 6.3 and 11 to 19 at the smaller `d/n` shapes. These are different shapes, comb-width choices and polynomial-phase shares (85-93% there, 0.5-17% here), not a matched comparison, and they are not offered as a correction of that note.

7. Deviations, limitations and what is unfinished

1. **The H3 gate wording.** "Any rho `S` outside [0.3, 4]" was meant for the per-size medians (the protocol's tables are medians) but was written without saying so; read per run it is tripped by 5 of 84 individual rho runs (rho's cost has a heavy lower tail: the lowest run is 0.035). The consequence the protocol attached was "nothing is ranked against rho". We report the rankings anyway, **conditional on the median reading**, and the rho-free quantities (ops against the formula `F`, fit B, section 4's Cheon exponents) do not depend on it. The slope gate (the other half) was met.
2. **ecbench integration.** The rho arm ran through `ecbench` (sealed session, audit with 84 of 84 exact replays). The Cheon arms did not: ecbench's registered methods take only the target and its workload identity has no field for leaked powers, so a Cheon method would need a workload-identity extension that changes the hashed workload record CI replays for every committed session. Not done; the driver rebuilds the same workloads with `Workload::on` and the workload ids were checked equal (84 of 84). Cheon rows are therefore not in an ecbench database or `ecbench compare` output.
3. **Isolation.** All runs are L0 (shared host, load 8 to 10). Operation counts stand; wall times in the raw records are descriptive only and are not used.
4. **Replay scope.** `replay-certificate.json` re-executes, in one fresh process, every Cheon run up to 48 bits and target 0 at 52 and 56 bits (cost), and the matched `rho.negation` on the same workloads with the session's algorithm seeds (same binary, same unit); the whole rho session was additionally replayed by ecbench (`--replay-all`, 84 of 84).
5. **Curves are unregistered** (`docs/curves/registry.json`): the ICV1 slugs and EC1 aliases exist only in `sessions/rho-20261022/plan.json`; prose names the curves by family and size. Registration needs `scripts/build_curve_registry.py` and is left for a follow-up.
6. **Shape tolerance.** Realised `log d / log n` is 0.121 to 0.143 (`minus`) and 0.198 to 0.222 (`plus`), not exactly P-224's; fit B uses each curve's `d`. `n` has `b - 0.42` bits for the size label `b` (`p` starts at `1.5 * 2^(b-1)`).
7. **Rho variance.** Six targets per curve; rho's per-run cost spans a factor of up to 39 within one curve (`plus` 48 bits) and `S` runs from 0.035 to 2.3, so per-size ratios are noisy (for example `minus` 40 bits reads 1.70 where rho's six runs range from 71,008 to 1,447,215). The pre-registered fits average `log2` per size; the bootstrap intervals include this.
8. **Not run:** the kangaroo variant, sizes above 56 bits (word size allows 62), `minus` curves for the `n + 1` algorithm and vice versa, a search over `d` for the best split, and Cheon in any non-prime-field group.
9. **Tests are in `tests/cheon_counted.rs`**, not in the module: in this sparse checkout the library's `cfg(test)` build cannot compile (its `include_str!` fixtures are not checked out), whereas an integration test links the release library.
10. **Measured binaries.** The measured Cheon binary was built from the sources at commit `4cbd0ed80`; later commits add only the test file, figure code and reports (the `run` path is unchanged).

8. Next improvements to try

1. Add a leaked-power workload (`aux_powers`) to ecbench so Cheon is a registered method with sealed, replayable sessions.
2. Price and shrink the `n + 1` polynomial phase (a Straus or Pippenger multi-scalar sum instead of 3 sums of naive double-and-add terms) and the comb baby/giant balance; both attack the constants above the formula.
3. A memory-lean kangaroo variant at 60 to 62 bits, and more targets for rho (the interval width is dominated by rho's variance).
4. Register the 14 curves and run the same shapes at `d` fractions between the two measured here to see the exponent in `d/n` directly.

9. Reproduce

```
export CARGO_TARGET_DIR=/home/user/crypto/target
cargo build --release --example cheon_p224_shape --example cheon_p224_shape_analyse --example cheon_p224_shape_validate
E=$CARGO_TARGET_DIR/release/examples
$E/cheon_p224_shape check --curves curves.json
ecbench run --wait --spec ecbench-spec.json --out sessions/rho-20261022 --cpus none --allow-busy
ecbench verify --dir sessions/rho-20261022 --replay-all --out sessions/rho-20261022.audit.json
$E/cheon_p224_shape run --curves curves.json --out runs/cheon-minus.jsonl --family minus
```

```
$E/cheon_p224_shape run --curves curves.json --out runs/cheon-plus.jsonl --family plus
$E/cheon_p224_shape xcheck --curves curves.json --max-bits 40 --targets 3
$E/cheon_p224_shape replay --curves curves.json --runs runs/cheon-all.jsonl --rho-records sessions/rho-20261022/records.jsonl --out replay-certificate.json
$E/cheon_p224_shape_analyse --curves curves.json --runs runs/cheon-all.jsonl --rho sessions/rho-20261022/records.jsonl --out results
$E/cheon_p224_shape_validate --curves curves.json --runs runs/cheon-all.jsonl --rho sessions/rho-20261022/records.jsonl --results results/results.json --out
validator-report.json
```

`curves.json` regenerates with `cheon_p224_shape gen --out curves.json` (about 80 s). Counts are deterministic: a rerun reproduces every `ops` value.

10. Files

`PROTOCOL.md`, `curves.json`, `ecbench-spec.json`, `runs/`, `sessions/`, `xcheck.txt`, `replay-certificate.json`, `validator-report.json`, `unit-tests.txt`, `results/` (tables, `results.json`, figures), `reports/run-20261022T0538Z/` (run report), `REPORT.md`/`REPORT.pdf` (this report), `render.sh`, `report.css`. A copy of the PDFs and graphs, and the validation-gate manifest `VALIDATION.json`, are in `benchmarks/20261022-cheon-p224-shape/`.